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SURVEY

Autonomous Mobile Robot Path Planning Techniques–A Review: Classical and Heuristic Techniques

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ABSTRACT Autonomous mobile robots (AMRs) are increasingly used in various applications, including transportation, logistics, and healthcare. One of the central challenges in AMR deployment is path planning, which involves finding an optimal and collision-free route from the robot's initial position to its destination. This review, which builds upon a two-part study, presents a comprehensive overview of state-of-the-art techniques for AMR path planning. This paper focuses on classical and heuristic-based strategies, providing valuable insights into their foundational roles in autonomous mobile robot navigation. We categorize existing approaches based on their core principles, including graph-based, heuristic-based, metaheuristic-based, and learning/reasoning-based methods. For each category, we analyze the strengths and limitations of the representative techniques and evaluate their effectiveness in various operational scenarios. Furthermore, we identify emerging research directions and ongoing challenges in AMR path planning, such as multi-robot coordination, adaptation to dynamic environments, and integration with human–robot interaction frameworks.

INDEX TERMS Autonomous mobile robots (AMRs), path planning, Dijkstra algorithm, A* algorithm, rapidly-exploring random tree (RRT), artificial potential field (APF), cell decomposition, heuristic methods, meta-heuristic algorithms, dynamic environments.

I. INTRODUCTION

Autonomous mobile robot (AMR) is a type of robot that can comprehend and navigate its surroundings on its own. AMRs use artificial intelligence, machine learning, and computing for path planning to interpret and navigate their environment. Suppose AMRs encounter an unexpected obstacle during navigation in the given environment. In that case, they use an intelligence-based embedded navigation technique to adjust their speed and either halt or choose an alternate route to bypass the obstacle, ensuring seamless navigation to perform their given task. AMRs have applications in a wide range of fields, including greenhouse environments [1], military bomb diffusion [2], agricultural operations [3], healthcare [4], education [4], Industry [5], security and safety [6], [7], optimal energy utilization [8], path-finding [9], etc.

There are several challenges associated with navigation within these environments, and to succeed in these environments, AMRs rely on their intelligence and ability to model the surroundings, localize themselves, control their mobility, and detect and avoid obstacles (both static and moving). One of the major challenges is path planning, which poses a significant challenge for these robots, as it directly influences their ability to navigate, avoid collisions, and complete tasks in a congested environment. Path planning, as one of the most extensively researched topics today,

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involves finding an optimal or nearly optimal route for a robot to navigate freely from a starting point to a desired endpoint in an environment with varying complexities. A good path planner can enhance a mobile robot's efficiency by reducing the time spent on tasks and the energy required for them. Path planning is a complex problem that has been proven to be an NP-hard problem [10], which means that finding an optimal solution in a reasonable period is highly difficult [11], [12]. However, Multi-Objective Evolutionary Algorithms (MOEAs) offer a promising solution to solving this problem efficiently. By considering multiple objectives simultaneously, MOEAs can generate a set of Pareto-optimal solutions that balance different trade-offs, such as energy efficiency, path safety, velocity smoothness, and route length. These objectives are crucial in creating effective and safe paths for robots or autonomous vehicles [11], [13], [14]. The path-planning problem has been the subject of extensive research, with researchers making significant contributions to the development of testing and planning methodologies. As the navigation requirements of AMRs become increasingly stringent, these contributions continue to expand and evolve with each passing day [15].

Considering the state of the art in the field, a wide range of approaches for path planning exists, making the selection of an ideal method a challenging task. The path planning process can be categorized into two main types based on the environment's accessibility: **Global Path Planning** and **Local Path Planning**. Identifying the type of path planning required before initiating the path planning procedure is important [16], [17], [18], [19]. Global path planning occurs offline before autonomous mobile robots (AMRs) are deployed in the environment. During this process, the robot determines the environment's topology, the locations of barriers, its starting and ending positions, and other relevant factors to construct an environmental map [17]. The method utilizes optimization criteria (e.g., distance, time, and energy) to determine the robot's optimal path to achieve its goal. However, global route planning fails when the environment or obstacle placements are uncertain or unfixed, rendering collision avoidance ineffective. Standard methods in this category include graph-based, probabilistic, and heuristic-based methods such as A* and Dijkstra. The choice of the algorithm depends on the particular requirements of the problem, such as the scale of the environment, the presence of dynamic impediments, and the desired trade-off between speed and optimality [20], [21], [22], [23], [24], [25]. On the other hand, local path planning, also known as trajectory planning or motion planning, finds a smooth and accident-free path for a robot to follow through a workspace while satisfying its dynamic constraints, such as velocity and acceleration. It utilizes intelligence to guide the robot to its desired destination while navigating obstacles. Unlike global path planning, it does not require prior knowledge of the environment and generates a path based on real-time sensor data. Local path planning continuously adapts and plans new pathways as new information becomes available using real-time data. The strategy is effective in a fast-paced setting but could be ineffective when the target is far away [26]. Some common approaches for this approach include swarm intelligence, sampling-based methods such as rapidly exploring random trees (RRTs), and optimization-based methods, such as genetic algorithms (GAs) [27], [28]. Combining the two methods of path planning (global and local) may be a viable solution to reap the benefits of each approach while mitigating their drawbacks [26], [29], [30].

Researchers and academics have proposed and utilized numerous strategies to address the significant challenges that mobile robots face during navigation, whether in a fixed or dynamic environment with static or moving obstacles. When planning the path for a mobile robot, it is essential to consider its configuration space. The configuration space refers to the set of all possible configurations the robot can take, including its position and orientation in space. In other words, the configuration space defines the robot's boundaries within [31]. Ensuring the configuration space is compatible with the path planning algorithm to simplify the problem is essential. This compatibility ensures the robot can navigate smoothly and safely through its environment while avoiding obstacles and other hazards, especially in crowded environments [28]. A multitude of reviews on AMR path planning strategies have been published over the years, including [21], [23], [32], [33], [34], [35], [36], [37], [38], among others. However, many of these works focus on narrow algorithm subsets or specific applications, often assuming static or structured environments. For example, the authors of [32] conducted a comprehensive survey on coverage path planning, offering valuable insights into environment mapping and coverage efficiency. While beneficial in structured settings, their work does not explicitly address comparative analyses across broader planning families or techniques for dynamic, real-time applications. Reviewers have categorized AMR path planning techniques into distinct groups. For Instance, Souissi et al. [39] proposed a hierarchical classification framework that distinguishes strategies based on physical constraints, determinism, environment modeling, and whether planning is performed online or offline. Other works classify approaches such as classical or reactive approaches [16], heuristics, AI, and bio-inspired methods [34], reactive computing, optimal control, c-space search, and soft computing [35], among others. Choosing the most suitable approach for path planning remains challenging, particularly under application-specific constraints [40].

During the initial phase of this work, we observed that existing surveys of AMR path-planning methods were incomplete, as they overlooked some commonly used approaches. Furthermore, some evaluations only identified a subset of strategies as primary, leaving a significant portion of the field unexplored. This survey on AMR path planning aims to conduct a comprehensive and methodical review, critically analyzing a spectrum of algorithms tailored for static and

dynamic settings, including scenarios featuring stationary and mobile obstacles. The survey aims to systematically categorize these algorithms based on their underlying principles, providing a detailed evaluation of their advantages, drawbacks, and application boundaries. This dual-faceted literature review aims to elucidate the most effective strategies for specific operational contexts, highlighting recent advancements and identifying avenues for further research and technological breakthroughs. Additionally, this study extends its analysis beyond conventional 2D environments to incorporate complex 3D navigation landscapes.

This paper thoroughly investigates classical and heuristic-based approaches commonly used in AMR path planning. Path-planning strategies like cellular decomposition, artificial potential fields, and others are reviewed, along with heuristic approaches such as Dijkstra and A* algorithms. Furthermore, we include a table that summarizes the applicability of these path-planning approaches in different environmental contexts, considering varying complexities, for easy reference. This study draws on a substantial corpus of literature, comprising journal articles and conference proceedings, to provide a detailed analysis and categorization of the approaches.

The remainder of the article is organized as follows: Section II mentions the AMR path-planning techniques and their categories. Section III presents the analysis and discussion. Section IV discusses the challenges and opportunities in AMR path planning, and Section V concludes the paper, detailing the prospects.

II. AUTONOMOUS MOBILE ROBOTS PATH PLANNING TECHNIQUES

The simplified AMR path planning algorithm classes are presented in Figure 1, along with several methods within each category. The techniques can be broadly classified into four groups: traditional methods, heuristic methods, meta-heuristic algorithms, and learning- and reasoning-based methods. This classification is based on the underlying ideas and methods of creating and returning a path. The "roadmap technique," "artificial potential fields," and "rapid random tree*," among others, are just a few examples of traditional methods. There are Dijkstra, A*, and D* algorithms in the category of heuristics. Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and other bio-inspired techniques fall under the meta-heuristics approach. Lastly, there are reasoning-based techniques, including deep reinforcement learning, fuzzy logic control, and neural network control, among others. In this phase of the work, our primary focus will be on addressing the first two categories of path planning. Subsequently, we will dedicate the next part of our work to exploring the remaining two categories.

A. CONVENTIONAL APPROACHES

Many attempts have been made to tackle the AMR path planning problem using traditional techniques, including cell decomposition, the roadmap technique, and potential field methods. However, these methods are limited in their fundamental formulation when dealing with unknown or dynamic environments or obstacles; they have only been utilized for the global path planning task, where the knowledge of the environment and the position of obstacles are fully known before navigation. They are also known to require a lot of calculation. Furthermore, researchers investigating the root of the findings discovered that some of these traditional path-planning approaches could be more effective when used in tandem with one another or in conjunction with intelligent optimization algorithms, such as PSO, ACO, and DRL, among others [41], [42], [43]. In this study, we will place a greater emphasis on those conventional algorithms that are highly used and have been enhanced through hybridization and/or other techniques. Suppose the reader needs to look up outdated traditional path-planning techniques, such as the free space approach and topological method. In that case, they can do so by navigating to previous literature surveys (see [23]). Research in [23] also presented the three fundamental principles of mobile robot global path planning: environmental modeling, optimization criteria, and a path search algorithm.

1) CELL DECOMPOSITION APPROACH

The cell decomposition (CD) technique partitions the configuration space or continuous workspace into disjoint sets or grids collectively called "cells," making it effortless to represent and manipulate the environment for path planning [44]. This method relies on a connectivity graph that outlines the configuration space structure, facilitating the transition from one cell to another towards the goal [45]. Typically rectangular or square, cells are defined by their unique, adjacent positioning and assigned a value based on characteristics such as obstacles or environmental factors. This valuation enables the calculation of the most efficient route through the grid, effectively mapping the robot's trajectory [46]. There are three basic types of cellular decomposition methods: exact, approximate, and probabilistic [32], [37], [47].

Exact CD (ECD) precisely divides the free configuration space into fully defined cells that do not overlap with one another. The space inside each cell can either be fully traversed or not [48]. It's particularly effective in environments with well-defined obstacles and boundaries. It requires an exact representation of the space, partitioning it into simple geometric shapes, such as polygons in 2D or polyhedra in 3D. These cells are then analyzed to determine their connectivity and navigability, allowing for the construction of a precise path-planning graph. Paths are generated by linking adjacent cells without obstacles, ensuring a straightforward route from the start to the goal [49], [50]. Despite its accuracy and suitability for a structured environment, ECD can be computationally demanding in complex areas. Figure 2 depicts a trapezoidal ECD method used in partitioning a space. This approach divides the

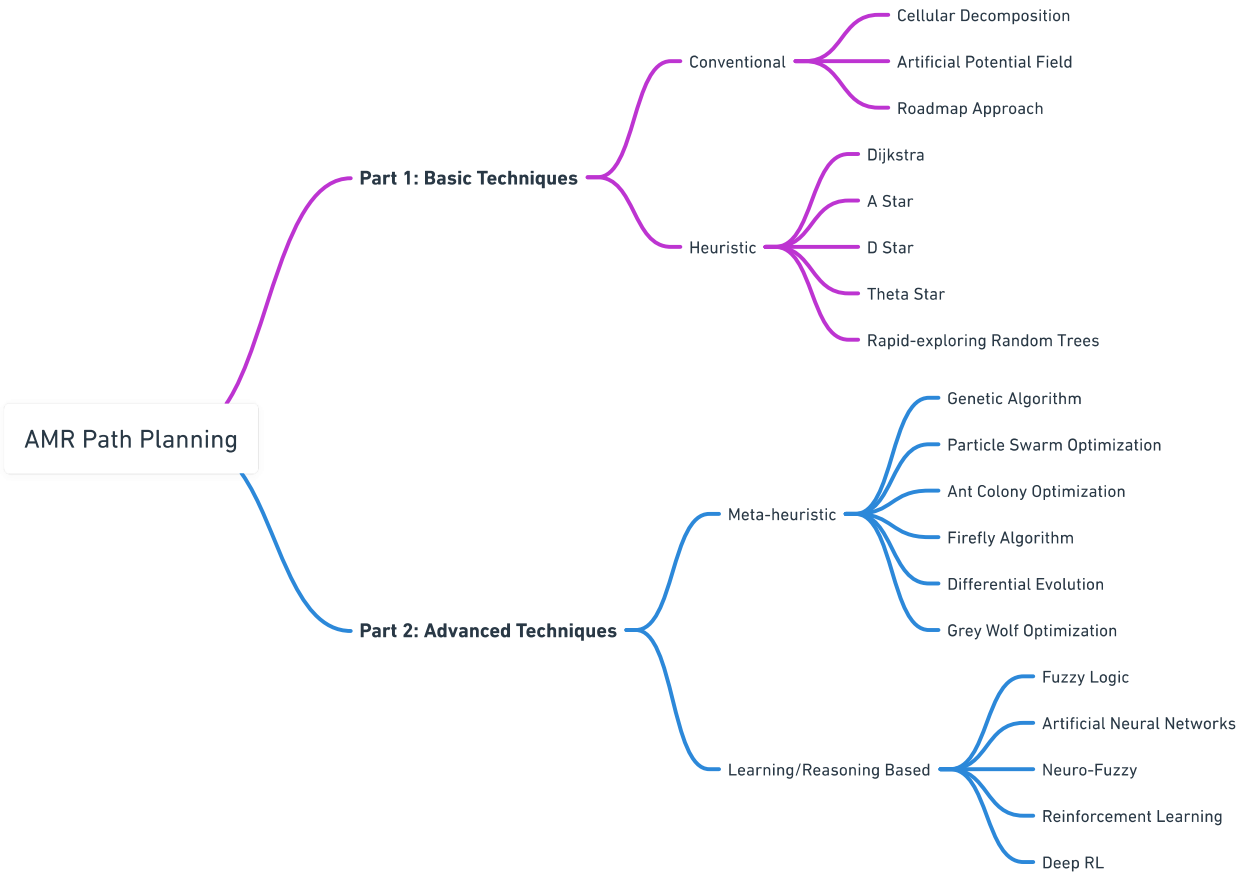


FIGURE 1. Classification of AMR Path Planning Techniques. The techniques are categorized into two main groups: basic and advanced. This paper primarily focuses on the basic techniques (highlighted in purple), while the advanced methods will be explored in a subsequent part of this study.

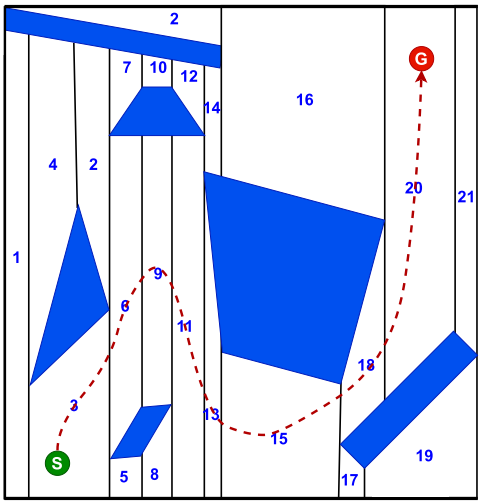


FIGURE 2. Trapezoidal cell decomposition.

environment into trapezoidal cells, with each cell numbered to facilitate navigation planning. Connecting cell centers and avoiding obstacles as necessary are the ways to build the path from the start (S) to the goal (G) [51].

Approximate CD (ACD) streamlines environment representation by organizing the configuration space into a cell grid that estimates navigability [52]. Unlike ECD, ACD does not require cells to match the shape of obstacles or free spaces precisely [48]. The presence of barriers categorizes cells while considering the possibility of partial occupancy. This method notably lessens computational demands, favoring environments with complex or irregular obstacles. In ACD, planning a path involves creating a connectivity graph that shows how cells can be traversed, considering both fully and partially free cells as navigable [53]. ACD balances efficiency and path quality, making it adaptable for various applications. Probabilistic CD (PCD) employs predefined cell shapes to model the environment, similar to ACD. One thing that makes PCD different is that the lines between cells do not always match up with real-world features. This differs from ACD, where cell divisions may more closely align with physical lines or obstacles [54]. Both PCD and ACD offer rapid implementation benefits, yet their effectiveness diminishes in environments where open spaces are minimal compared to the overall area. The survey in [55] and [56] discusses and explains additional CD techniques.

A significant challenge with CD techniques is their oscillations in the presence of barriers and constricting pathways [57]. The phenomenon has been noted in the literature, suggesting a need for refined algorithms that

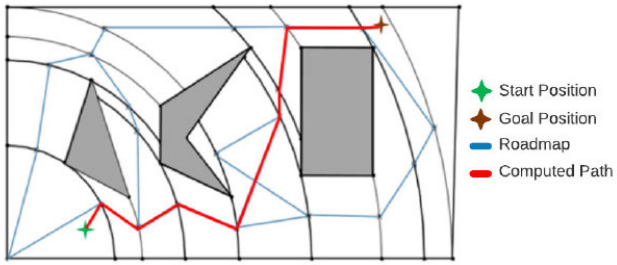


FIGURE 3. Route generated by RCD [59].

	Coordinates		VCD Alg.		RCD Alg.		improvement
	Start	Goal	Length	Time (s)	Length	Time (s)	in length %
1	0,0	10.0 , 7.5	15.61	0.45	15.54	0.34	0.44 %
2	0,0	10.0 , 7.5	17.88	0.46	16.66	0.43	6.8 %
3	0,0	10.0 , 7.5	21.38	0.43	17.37	0.41	17.2 %
4	0,0	10.0 , 7.5	19.55	0.43	15.88	0.43	18.7 %

FIGURE 4. Comparison between RCD and VCD regarding the path length and processing time [59].

mitigate these effects. The CD has been frequently employed in robot path planning, as evidenced in previous studies [46], [51], [58], [59], [60]. Reference [51] introduced a novel method for navigating mobile robots in complex environments. This approach integrated ECD and RRT algorithms, creating an adaptable and efficient motion planner. The proposed technique comprised two layers: the first utilized trapezoidal CD to identify open areas and feasible paths in the workspace, while the second employed RRT to determine the optimal route in the modified environment. The authors in [58] conducted a quantitative analysis of CD techniques and cost functions for mobile agent path planning in the presence of strategically positioned obstacles, addressing optimization and computational complexity. During the quest for a workable solution, a sensing-based online CD approach was proposed in [61] to enable AMRs' adaptive coverage of non-rectilinear environments. Salama et al. in [59] introduced radial CD (RCD) as an improved CD for optimal path planning in obstacle-rich settings, outperforming vertical CD (VCD) in terms of path length, as highlighted in Table 4. RCD efficiently navigates complex, non-rectilinear environments, offering practical implementation benefits. Fig. 3 depicts a typical route generated using RCD, showcasing its ability to produce shorter paths with reduced computational time.

The research in [62] introduces an expanded Douglas-Peucker polygonal approximation technique tailored for path planning with curvilinear obstacles. This method enhances the CD capabilities, providing improved solutions for scenarios involving curved obstacles. The study in [60] developed a new algorithm based on approximate CD for efficient area coverage of mobile robots. This algorithm enables real-time updates to the coverage plan, allowing the robot to adapt to environmental changes as it explores. The authors in [63] introduced a groundbreaking method in their research on effective path planning for AMRs. This method decomposes co-safe Linear Temporal Logic (LTL) specifications into independent tasks tailored for 3D environments. This approach, which leverages a novel recursive CD algorithm and an optimization framework for task assignment, enables drones to execute complex, distributed missions, significantly enhancing path-planning efficiency and autonomy. Recent advancements in cell decomposition techniques have significantly improved autonomous mobile robot path planning across various applications. The research in [64] tackled multi-objective optimization in obstacle-rich environments using a cell mapping algorithm, balancing distance, navigation time, and control effort. Reference [65] refined Boustrophedon Cellular Decomposition (BCD) for multi-robot indoor coverage, introducing improved segmentation and task allocation to enhance efficiency. The authors in [66] leveraged CD within a reinforcement learning framework for UAV path planning, reducing environmental complexity and improving obstacle avoidance. Reference [67] introduced a centroid-based CD method that combines a bio-inspired approach with graph-based mapping and ant colony optimization to refine trajectory planning and enhance navigation efficiency in complex environments. Meanwhile, [68] employed a Quadtree-based CD to optimize RRT, prioritizing safer, larger cells to enhance sampling efficiency and navigation in cluttered terrains. These contributions collectively demonstrate the versatility of cell decomposition in enabling efficient and adaptive path-planning strategies.

2) ROADMAP APPROACH

The roadmap or highway approach (RA) involves generating a network graph of feasible routes within a given space, considering both available and occupied spaces. The primary objective is to establish a network of collision-free and safe pathways from the mobile robot's initial position to its destination. Essentially, this approach condenses the robot's configuration field, encompassing all possible movements, into a network of one-dimensional lines akin to a graph search process [69]. The four steps that outline the general roadmap approach for path planning include map representation, graph generation, path planning, and trajectory generation.

The best-known road maps are the Voronoi diagram (VD) and the visibility graph (VG). The visibility graph approach is a graph-based method for path planning that represents the environment as a set of nodes and edges [70]. Nodes are defined as points in the environment, and edges are defined as straight lines connecting two nodes [47]. The approach works by finding the shortest path between two graph nodes that do not intersect any obstacles. This method is only helpful in environments where the robot can move in straight lines and the environment is free of narrow passages. On the contrary, the Voronoi roadmap approach uses Voronoi diagrams to represent the environment. In this approach, the environment is classified into a set of Voronoi cells, each representing a region of space closest to a single node. The paths a

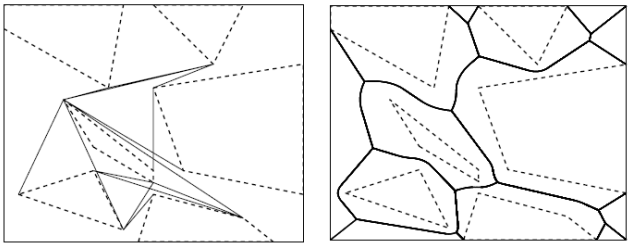


FIGURE 5. Roadmap obtained from visibility graph (left) and Voronoi diagram (right) [72].

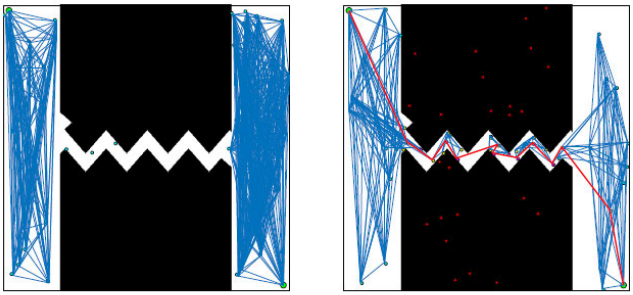


FIGURE 6. Comparison between traditional PRM and hybrid PSO-PRM [76].

robot can take between the nodes are defined by their edges. This approach benefits environments with narrow passages or obstacles that are difficult to navigate around [71]. Roads in the Voronoi diagram stay as far away from obstacles as possible, resulting in a tighter path that is slightly longer than the visibility graph [47], [57]. Figure 5 gives a typical road map. It was created from a graph with reduced visibility and includes all obstacle edges not in contact with the configuration space boundary. In contrast, the roadmap on the right was created using a Voronoi diagram, where the configuration space boundary is viewed as a barrier.

Many academics have employed the standard roadmap technique to tackle the static path-planning problem. Furthermore, advancements have been made to extend the capabilities of this technique in a dynamic setting with many types of considerations through upgrades and hybridization; some of the most recent successes in this respect have been noted in this review study [73], [74], [75], [76], [77]. To reduce the computational time required by the VG approach, [73] introduced a technique where small obstacles are grouped, and the operating environment is split for parallel processing. In a separate study [74], a novel algorithm for path planning of AMRs in dynamic environments was introduced. The algorithm utilizes Voronoi diagrams and computational geometry to classify moving objects based on their positions, rates, and directions of motion. This classification enables the algorithm to assess the risk these objects pose to the vehicle's kinematics and select the most suitable action. The authors in [78] introduced a novel approach for generating optimal trajectories for robotic systems under static polygonal constraints. The method employs a model-based strategy to develop a dynamic model of both the system and the constraints. This enables trajectory optimization that minimizes system costs while satisfying the imposed restrictions. The technique demonstrates efficient performance and exhibits viability and adaptability despite diverse environmental and modelling challenges. The primary drawback of the method is its substantial computational demands. Reference [79] introduced the Morphological Dilation VD Roadmap (MVDRM) algorithm to enhance roadmap-based path planning for AMRs. By integrating morphological dilation with VDs, their method enhances path safety and reduces computational complexity, outperforming PRM and conventional VDs in terms of efficiency and obstacle avoidance. The importance of optimizing both accuracy and computational efficiency cannot be neglected in path planning for AMR. Hence, the article [80] presents a recently introduced path planning method named the Fast Path Planner (FPS), which relies on a sparse visibility graph and bidirectional breadth-first (bb-f) search to generate optimal paths efficiently. The approach utilizes the V-graph to significantly accelerate computing by reducing the number of nodes and edges in the graph, as well as the bb-f search. This allows the FPS to simultaneously search from the start and goal nodes, allowing the algorithm to swiftly settle on a path solution.

The probability roadmap (PRM) is a notable extension of traditional roadmap methods, offering a probabilistic framework for path planning in complex environments. Unlike deterministic approaches like the VG and VD, PRM employs a randomized sampling strategy to generate nodes within the configuration space [81]. By randomly sampling feasible configurations and connecting them through local planners, PRM constructs a network that captures the connectivity of the free space, effectively representing the topology of the configuration space [82]. This probabilistic nature grants PRM robustness in handling dynamic and uncertain environments, as it can adaptively explore the space without relying on explicit geometric structures. Furthermore, PRM facilitates efficient querying for collision-free paths through its pre-computed roadmap, enabling real-time path planning in dynamic scenarios [83]. Advancements in PRM include techniques for improving sampling efficiency [76], [84] and addressing high-dimensional spaces through dimensionality reduction methods [85]. Figure 6 compares the performance of the enhanced PRM with traditional PRM, showing its superior capability in narrow environments. Despite its versatility and effectiveness, PRM may encounter challenges in scalability as workspace complexity and dimensionality increase. Nevertheless, ongoing research efforts continue to refine and extend PRM-based approaches, making them indispensable tools in AMR path planning.

3) ARTIFICIAL POTENTIAL FIELD APPROACH

The Artificial Potential Field (APF) approach has been extensively utilized in robotics for path planning and collision

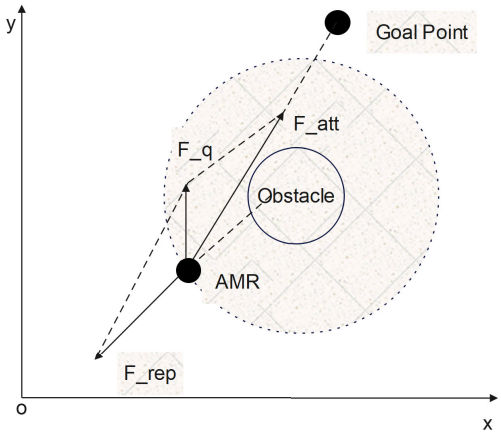


FIGURE 7. Force analysis of AMR using traditional APF.

avoidance. It creates a virtual field around the robot to guide its behavior based on desired outcomes [86]. This technique employs potential energy to generate a force field that attracts or repels the robot toward or away from specific environmental points. By adjusting the shape and strength of the potential field, the robot can navigate toward goals, avoid obstacles, and accomplish predefined objectives. The potential field consists of two types: "attractive," which pulls the robot towards the goal, and "repulsive," which pushes it away from obstacles [34]. The strength of these fields varies depending on the distance between the robot and the goal or obstacle. By computing the gradient of the potential field, the robot can determine the direction of the goal and navigate towards it accordingly [57].

Figure 7 depicts the fundamental concept of APF, where obstacles in the environment exert repulsive potential while the robot's destination exerts attractive potential. The autonomous robot's path is determined based on the gradient of the total artificial potential. Different mathematical formulations have been used to define the artificial potential fields, and several methodologies for leveraging the gradient of the total potential to discover a path for the robot have been proposed [41], [87].

The attractive force (F_{att}) is given by

$$F_{\text{att}} = \frac{1}{2} * C_{\text{a}} \rho(\text{ goal })^2 \tag{1}$$

The repulsive force (F_{rep}) is given by

$$F_{\text{rep}} = \frac{1}{2} * C_{\text{r}} * (1/\rho(\text{q}) - 1/\rho_0)^2 \tag{2}$$

where C_{a} and C_{r} are positive coefficients, ρ_0 is the potential factor.

Hence, the resultant force is given by

$$F_{\text{q}} = F_{\text{att}}(q) + F_{\text{rep}}(q) \tag{3}$$

This method undoubtedly relies on a technique where the map is predetermined. The APF approach has been widely employed and refined over the years to resolve challenges associated with path planning for mobile robots. This review highlights recent advances in using APF techniques for real-time path planning in dynamic environments. The approach discussed in [88] addresses the challenges that mobile robots encounter in static and dynamic complex environments. Researchers applied PSO for optimal parameter selection in the PID controller, enhancing robot navigation, and utilized a modified APF algorithm to determine collision-free paths. The study in [89] introduces a novel APF method that accounts for dynamic barriers and provides efficient solutions regarding path length and algorithm runtime. The method's strengths include the creation of a discrete potential field, the integration of heuristic-based rules to mitigate local minima, and enhanced obstacle avoidance. Its near-real-time performance renders it suitable for practical applications. For effective path planning and replanning as necessary, the research in [90] presents an improved APF strategy in conjunction with a map augmentation method, thereby preventing the endless loop problem experienced by mobile robots when using the standard APF method.

In reference [91], an improved APF method was introduced to tackle challenges in dynamic environments. This approach creates collision-free paths using repulsive force fields and attractive potential functions for destination and team formation. It addresses issues such as latency, local minima, and unreachable goals commonly associated with traditional APF. The research in [92] introduces the Safe APF (SAPF), an enhanced iteration of the APF algorithm designed to overcome path-planning challenges by incorporating a versatile obstacle potential and an analytical tuning procedure. SAPF is recognized for its capability to evade simple local minima, maintain safe distances from obstacles, and navigate narrow corridors. APF has also been used in conjunction with AI techniques, such as fuzzy logic and neural network control [93], [94], as well as Q-learning and reinforcement learning (RL) [95], [96]. Over time, more significant outcomes have been realized.

Recent advancements in APF-based methods have significantly improved AMR path planning by addressing issues with local minima, optimizing trajectory smoothness, and integrating hybrid approaches. The study in [97] introduced a bidirectional APF-based ant colony optimization (BAPFACO) algorithm to enhance pathfinding efficiency by improving convergence speed, reducing the number of turning points, and overcoming local optima. Similarly, [94] proposed a fuzzy A-based quantum multi-stage Q-learning APF model, integrating quantum computation and reinforcement learning to adapt to obstacle changes while reducing computational costs dynamically. The work in [98] presented an improved APF method with a dual-loop trajectory tracking control strategy, enhancing obstacle avoidance and trajectory stability for AMRs. Meanwhile, [99] developed an APF-enhanced multi-objective snake optimization (APF-IMOSO) algorithm, which balances safety, energy efficiency, and path length through meta-heuristic improvements. Addressing local minima challenges, [100] introduced a subarea-based

APF model that refines local navigation strategies by dynamically adjusting repulsive forces to create smoother and more efficient paths. Finally, [101] proposed a fusion of A* and APF for dynamic path planning, leveraging symmetry constraints and Bézier curve optimization to ensure trajectory efficiency and obstacle avoidance. These contributions collectively showcase the adaptability of APF in improving real-time path planning for autonomous mobile robots, particularly in complex environments.

B. HEURISTIC APPROACH

A heuristic is a practical method for swiftly and intelligently traversing the solution space to find an exact solution in a reasonable time frame. Heuristic algorithms are regarded as problem-dependent or graph-based methods, mainly used to select the nearest neighbor [102]. In path planning, heuristic algorithms use rules of thumb or experience to find a feasible solution that may not be optimal. Some of the algorithms in this category include A* search, Dijkstra's algorithm, and the Any-angle algorithm. In the subsequent sections, recent studies in this area have been discussed.

1) DIJKSTRA ALGORITHM (DA)

Edsger W. Dijkstra introduced the Dijkstra algorithm in 1959 as a fundamental graph-based path-planning technique [103]. This algorithm identifies the shortest path between two connected nodes by exploring all potential paths using a priority queue, prioritizing paths with the lowest cumulative cost [104]. It is particularly effective in simple environments where the optimal route is evident. Figure 8 illustrates Dijkstra's method using Euclidean distances: Shortest paths are marked in red, while blue lines indicate edge relaxation, where the estimated path costs are updated [105].

Dijkstra's algorithm is widely employed for global path planning in predetermined and static environments. It is also adaptable to local path planning and dynamic settings. However, the method can be inefficient, as it may require scanning almost the entire map to identify the shortest route. This process can become time-consuming and computationally intensive, particularly with a large number of nodes [58]. The traditional DA, a greedy search technique, focuses solely on identifying the shortest path without considering its viability or practicality for real-world applications. This limitation makes it less suitable when cost, time, and effort are critical factors [106]. A modified Dijkstra algorithm introduces confidence intervals on each graph edge, enhancing the conventional approach by providing insights into the level of navigational freedom and mitigating computational constraints. This adaptation enables the practical exploration of possible routes, accommodating larger, cost-intensive graphs by identifying the shortest path with minimized stress [107]. In a separate adaptation, a modified DA is implemented in the V-REP open-source environment to address limitations in mobile devices navigating areas with static obstacles.

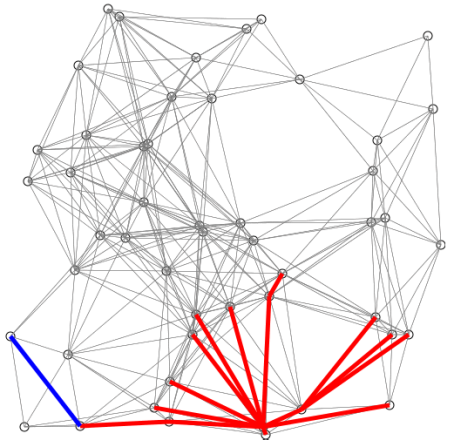


FIGURE 8. Dijkstra's path planning method on a graph.

This version enhances efficiency by incorporating a heuristic function, thereby reducing the time and energy required to reach the target [108].

Researchers have enhanced the traditional DA by incorporating heuristics and pre-processing techniques to reduce the frequency of truck movements in warehouse path planning applications [109]. Similarly, an improved version of Dijkstra's algorithm has been developed to enhance the efficiency of mobile robots, featuring collision avoidance capabilities. This method involves equipping a DIY mobile robot with a range sensor and a controller for motion and collision detection. The environment is modeled as a digraph with vertices and weighted edges representing accessibility. The obstacle avoidance mechanism utilizes these weights to guide the path planning process, ensuring the selection of the shortest and safest route to the destination [110]. A hybrid path planning method for mobile robots in changing environments is addressed in [111]. This method utilizes offline global optimization with the ABC algorithm to predefine control path points. Online, the system integrates these points into a graph, adding real-time updates for start and destination, and then applies Dijkstra's algorithm to find the shortest path. The approach adapts to environmental changes and is validated through MATLAB/Simulink simulations, accompanied by visual materials.

The DA remains a cornerstone in AMR path planning, with recent research focusing on improving its adaptability, efficiency, and integration with hybrid approaches. Research in [112] enhanced Dijkstra-based path planning for AMRs in the Robot Operating System (ROS) environments, leveraging LIDAR mapping and real-time localization to improve accuracy and performance. In complex indoor settings, the authors in [113] proposed a local path planning fusion of Dijkstra and PID control for Mecanum-wheeled robots, achieving better navigation through tight spaces and reducing completion time compared to conventional TEB algorithms. Further optimization efforts are presented in [114], where a weight-controlled particle swarm-optimized

DA (WCPSODA) is introduced to enhance route planning for autonomous robotic vehicles, demonstrating superior collision avoidance and efficiency. Building on hybrid strategies, the research in [115] combined Quantum-behaved PSO with deep RL (RL-QPSO Net), significantly improving global path optimization and real-time adaptability beyond traditional Dijkstra-based techniques. The work in [116] focused on energy-efficient path planning, comparing Dijkstra and A* algorithms and minimizing unnecessary turns and visited nodes to reduce energy consumption in grid-based environments. In dynamic industrial scenarios, the research in [117] proposed an improved Dijkstra framework integrated with real-time image processing and Canny edge detection for warehouse robots, resulting in faster computation and more accurate obstacle avoidance. Finally, [118] demonstrated the application of Dijkstra's algorithm on a service robot using ROS and ESP32 controllers, achieving precise navigation with minimal error in real-world trials. These developments collectively highlight the versatility and ongoing evolution of the DA in addressing modern path-planning challenges for autonomous mobile robots.

2) A STAR APPROACH

The A* algorithm is a method for navigating a grid and exploring pathways based on search principles and is known for its thoroughness, optimality, and efficiency [119]. It is commonly used in robotics, video games, and other areas where finding the most efficient path between two points is necessary [120]. An exhaustive algorithm seeks to discover all possible solutions to a problem, whereas an optimal algorithm aims to identify the solution with the least cost. The A* algorithm searches along the goal point while considering the cost of reaching each neighboring node and an estimate of the remaining cost to reach the target node.

The A* algorithm is considered an improvement over other search algorithms, such as Dijkstra's algorithm, as it is more efficient and can find the shortest path more quickly. A critical aspect of the A* methodology is formulating an appropriate evaluation function, and its quality heavily influences the algorithm's effectiveness [121]. The function takes the following form:

$$f_n = g_n + h_n \tag{4}$$

Here, g_n represents the most economical route from the initial node to node n, while h_n corresponds to an empirical estimation of the node's worth.

The A* algorithm, however, has some noticeable limitations, such as low search efficiency, the need to traverse many nodes, unnecessary path slanting through obstacles, limited node expansion direction, and unnecessary bends [122], [123]. Previously, the A* algorithm was frequently utilized for static or all-encompassing route planning problems, where the complete surroundings were stored or understood ahead of navigation. However, since the development of AI and nature-inspired algorithms, the method has been improved and combined to address various path-planning scenarios, such as those involving changing environments, moving targets, and 3D environments [124], [125]. In [126], an improved A* algorithm was developed to boost AMRs' search efficacy, path reliability, and ability to realize efficient route planning. A real-time AMR path planning procedure was proposed in [127]. The improved real-time A* algorithm was implemented in a grid map setup within the Khepera-II universe, which consisted of a quadrant-based environment populated with both stationary and moving objects. The planner effectively determines the optimum path by avoiding potential impediments. In [128], to figure out the best path for a robot in motion while avoiding collisions and obstacles, a memory-efficient A* grid-based route planning method was presented. The approach outperforms several previous grid-based optimization algorithms by identifying shorter paths requiring less effort and memory.

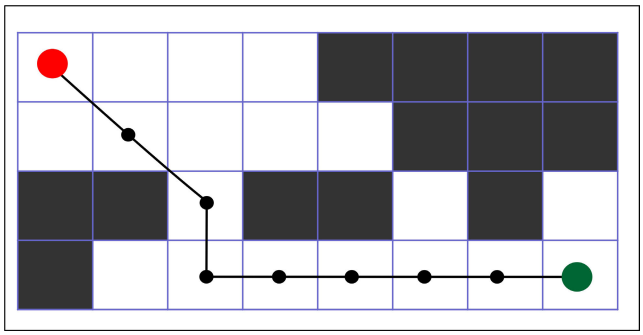


FIGURE 9. A sample path generated by the A* algorithm.

An online robot path planning method that combines the modified A* algorithm with an improved dynamic window (DW) technique was proposed to aid in planning problems in environments with random obstacles [129]. The A* algorithm's efficiency was enhanced by optimizing its heuristic function, search point selection method, and window approach for real-time routing. Additionally, a preview deviation angle tracking method successfully captured dynamic goal points. Simulation results suggest this approach is viable for local and global path planning. The dynamic window approach has also been used in [130], [131], and [132] to enhance the A* algorithm by smoothing the path, reducing search time, and minimizing excessive node traversal. The authors in [133] proposed a multi-objective path planning approach using an enhanced A* algorithm, along with a greedy algorithm. The A* algorithm's convergence is expedited by eliminating unnecessary nodes and modifying the evaluation function. Unlike the traditional A* algorithm, which is only helpful in locating a single goal point, combining the modified A* algorithm and the greedy algorithm is a viable solution for AMR multi-objective route planning.

To address the issues of redundant search nodes and prolonged search times, [134] developed a five-neighborhood A* algorithm, incorporating dynamic weights in the heuristic function and a second-order Bezier curve for path smoothing.

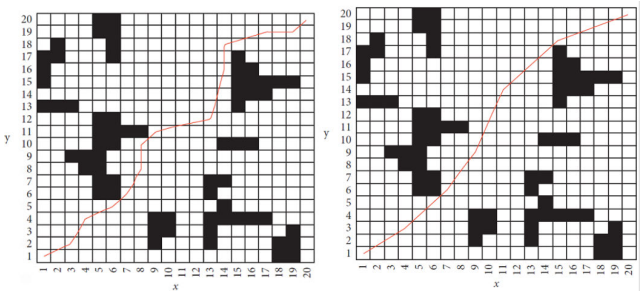


FIGURE 10. Path with classical and improved A* planning [129].

This algorithm achieved an 86% reduction in redundancy and search time. Zhang et al. in [135] also used dynamically adjusted weights in their A* algorithm to cut down on turns that were not needed and closeness to obstacles. Additionally, alternative enhancements to the A* algorithm have been proposed. Reference [136] introduced a bidirectional alternating search (BAS) strategy with an exponentially attenuated weighted heuristic function. Meanwhile, [137] proposed an advanced A* algorithm that integrates a tracking control strategy based on model predictive control theory. Research in [138] presents an improved algorithm that incorporates an adaptive environment information cost function, thereby enhancing the priority of neighborhood nodes and the key node selection strategy.

3) D* ALGORITHM

The D* approach is a dynamic variation of the A* method that helps the robot determine the best course of action in unpredictable settings [139]. D* is more favorable in dynamic environments, especially as the map size and complexity of the planning process increase [140]. The approach's problem space is formulated as a collection of states, where each state represents the direction in which the robot moves. There are n-by-n grids dividing up the area of exploration. Each cell in the grid has a state that indicates whether barriers are present or not. An optimized variant of the D* algorithm, called the D* Lite algorithm, has been applied in mobile robots to achieve enhanced search efficiency and improve robot safety, particularly in dynamic environments or in the presence of dynamic obstacles [141], [142]. Similarly, an extension of the D* algorithm called Focused D* [143] has been developed. It focuses on repairing paths created in traditional D* to significantly reduce the time required for initial path calculation and subsequent replanning. It has been extensively applied to planning and replanning in dynamic environments [144], [145].

The researchers in [146] proposed a hybrid algorithm that successfully combined the benefits of the D* heuristic approach and the PSO technique to address the route problems of AMRs in a dynamic environment. The D* obtains the optimal path, while the PSO ensures the path's resilience and optimality. The PSO helps smooth out the route's jagged edges. Similarly, in [147], a hybrid path-planning strategy for mobile robots in a dynamic world was proposed. The method benefited from particle swarm optimization and a better dynamic A* algorithm. The improved D* approach evaluates and predicts the cost of each node from the destination node. At the same time, the PSO identifies and presents the ideal path in an evolving environment with dynamic barriers. According to simulation results, the proposed approach performs efficiently, allowing the robot to reach the goal point without colliding and with minimal energy and time consumption. Authors in [148] suggested a hybrid algorithm for mobile robot path planning that effectively tracks dynamic goals and avoids introducing emotional barriers by combining the benefits of an enhanced APF with the D* algorithm. The modified APF was used to direct the robot along its course toward the dynamic target and away from the dynamic impediments. At the same time, the D* algorithm's cost function was incorporated into the attracting potential equations to enhance the algorithm's resilience to dynamic environments and objectives.

D* Lite can efficiently respond to environmental changes without recomputing the path. An incremental heuristic search is used for this path re-planning [144]. It reuses information from previous search iterations to update the path. This, in turn, reduces the computation time [149]. The D* Lite has been extensively used in various environments for path planning, including mixed environments with dynamic obstacles [141] and unknown environments [150], [151]. An optimized D* Lite algorithm was developed in [152] by introducing reverse search connection optimization after planning a path, thereby addressing the issue of robot safety not being prioritized and excessive turning times. It demonstrated a considerable increase in the efficiency and safety of robot movement. Another variant of the D* algorithm is the Field D* algorithm, proposed by D. Ferguson and A. Stentz [153], which generates smooth paths in the path planning process by introducing linear interpolation.

4) THETA* ALGORITHM

The Theta* algorithm is a type of any-angle path planner. It is an optimized version of the A* path-finding technique, commonly used to find the shortest path between two points in a graph. It employs a line-of-sight check to minimize the number of nodes expanded during the search process, resulting in faster and more efficient pathfinding [154]. Unlike A*, which requires a node's parent to be its immediate predecessor, the Theta* algorithm allows any previously expanded node in the search space to act as a parent for a given node [155]. The introduction of the line-of-sight (LOS) detection feature aims to reduce unnecessary node expansion by verifying whether a node's neighboring offspring and its parent are in a straight line, i.e., within the line of sight. Therefore, the path determined by the Theta* algorithm is not simply a collection of neighboring nodes but a sequence of line-of-sight nodes that are not constrained to a

grid-like structure. Through a cost evaluation process, Theta* approximates the length of the shortest path [156].

Initially devised as an improvement upon Theta*, Basic-Theta* emerged as a more computationally efficient variant, striking a favorable balance between path length optimization and runtime demands [157]. Comparative evaluations underscore its proficiency in optimizing shortest path outcomes [158]. An extension called Incremental Phi* [159] achieved expedited computational performance, accelerating Basic-Theta* by an order of magnitude by adopting a free-space assumption. The pseudo-code for the Theta* path planning algorithm can be found in Fig. 1. The research in [160] proposed a 3D any-angle path planning method based on improved Theta* and Lazy Theta. Both algorithms improve non-uniform costs by promoting smoothness and safety. In [161], the effectiveness of Theta*- RRT and GRIPS for kinematically deciding the best path for differential robots is compared. According to simulation and experimental results, Theta*-RRT is more stable and has a higher success rate than GRIPS, which is faster on broader synthetic terrain and generates relatively shorter routes. In addition, [162] introduces an improved version of the Theta* algorithm that addresses the computational limitations of the traditional Theta* algorithm, thereby making it more practical for real-world applications. The modifications include a dynamic weighting strategy and a revised distance calculation formula, which reduce computation time and ensure the generation of fast and smooth optimal global paths. The proposed algorithm outperforms the traditional Theta* and A* algorithms in terms of speed and memory usage.

An adaptive version of Theta* called MS-W-Theta* was developed in [163] to better handle real-world mobile robot applications, particularly in 3D environments such as logistics workshops. It implements an obstacle buffer in the 3D map of the environment, then utilizes an adaptive heuristic function to optimize the algorithm's cost function. Subsequently, it mitigates unstable speed or jitter phenomena, which commonly occur during operation, by smoothing the actual path using a Minimum Snap (MS) polynomial algorithm. Research in [164] combines the Theta* algorithm with a Timed Elastic Band (TEB) algorithm implementation on a wheel-legged mobile robot to dynamically plan a path for both the body and the wheel of the robot separately, classifying obstacles as either complete or incomplete. This was achieved using a hierarchical planning mechanism and multiple optimization algorithms. This implementation can be applied to more scenarios and has better performance, less computational time, and a shorter path length. Reference [165] extended the Theta* algorithm for fixed-wing UAVs by integrating clothoid curves to smooth waypoint transitions. Using Euler spirals improves trajectory continuity, reduces discontinuities, and respects flight dynamics. Adding adaptive exploration distance and a vision cone lowers computational complexity, significantly enhancing path smoothness and mission efficiency compared to traditional Theta* algorithms. Similarly, in [166], a hybrid Theta*-APF method was employed to address the issues of low efficiency, difficulty in handling obstacles, and numerous turning points in the global path planning of uncrewed aerial vehicles (UAVs).

Algorithm 1 Theta* Algorithm

Input: Start point s_{start}, Goal point s_{goal}, Map *map*
Output: Planning path
Initialize open set $\{open := \emptyset\}$ and closed set $\{closed := \emptyset\}$
Set $g(s_{start}) := 0$, $parent(s_{start}) := s_{start}$
open.insert(s_{start}, $g(s_{start}) + h(s_{start})$)
while $open \neq \emptyset$ **do**
  $s \leftarrow open.pop()$
  if $s = s_{goal}$ **then**
    return Path
  end if
  $closed := closed \cup \{s\}$
  for all $s' \in neighbors(s)$ **do**
    if $s' \notin closed$ **then**
      if $s' \notin open$ **then**
        $g(s') := \infty$
        $parent(s') := NULL$
      end if
      $g_{old} := g(s')$
      if LineOfSight($parent(s), s'$) **then**
        $Cost := g(parent(s)) + cost(parent(s), s')$
        if $Cost < g(s')$ **then**
          $g(parent(s')) := parent(s)$
          $g(s') := g(parent(s)) + cost(parent(s), s')$
        end if
      else
        if $g(s) + cost(s, s') < g(s')$ **then**
          $parent(s') := s$
          $g(s') := g(s) + cost(s, s')$
        end if
      end if
      if $g(s') < g_{old}$ **then**
        if $s' \in open$ **then**
          $open.remove(s')$
        end if
        $open.insert(s', g(s') + h(s'))$
      end if
    end if
  end for
end while

5) RAPID-EXPLORING RANDOM TREES ALGORITHM

The Rapidly-Exploring Random Trees (RRT) algorithm, devised by LaValle and Kuffner [167], is a renowned approach in sampling-based and graph-based strategies for autonomous mobile robot (AMR) navigation in various settings. RRT efficiently explores search spaces by generating

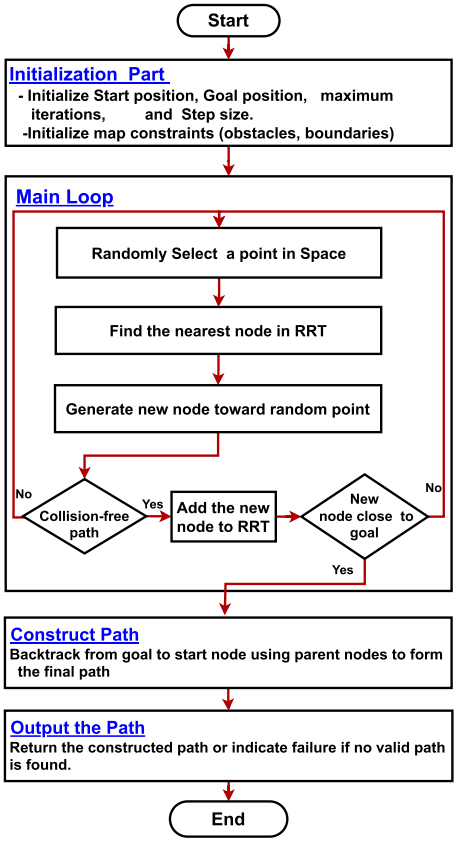


FIGURE 11. A simple flowchart of the conventional RRT algorithm for AMR path planning.

random samples that connect to form a tree-like structure [168], [169]. Sampling-based methods, such as RRT, offer computational advantages over search-based methods, particularly as the state space size and dimensions expand [170].

The RRT algorithm constructs a tree in the state space by randomly selecting nodes and connecting them to their nearest neighbors [81]. This process iterates, prioritizing unexplored regions to explore the space swiftly. One path progresses towards the goal while others navigate other areas. The goal is to find a feasible path from one point to another, avoiding obstacles and seeking an optimal solution. RRT* improves upon RRT by optimizing the path-finding process [83]. It searches within a specified radius to connect nodes, balancing path length and structure. This modification yields an optimal path to the goal and enhances adaptability to the environment [23], [171]. Figure 11 depicts a simple flowchart for the RRT algorithm for mobile robot path planning.

One of the key advantages of the RRT algorithm lies in its ability to adapt to dynamic environments, ensuring probability completeness and seamless scalability [169], [172], [173]. Through continual expansion of the tree structure using random nodes, the algorithm dynamically adjusts to environmental changes, making it suitable for real-world scenarios characterized by unpredictability or constant flux. However, RRT may encounter prolonged search times due to

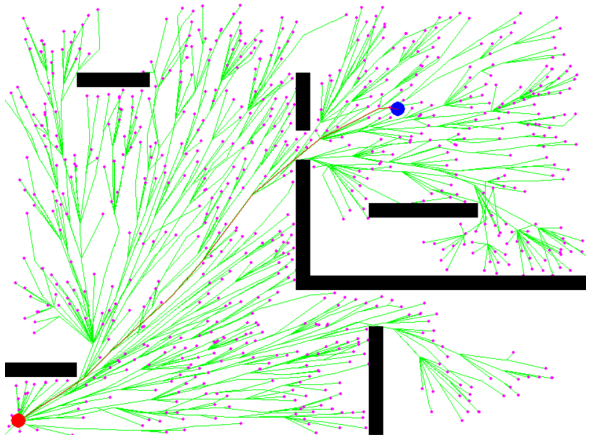


FIGURE 12. Discovering a viable path from start to goal using the RRT* algorithm [178].

its inherent randomness in sampling [174], [175], making it computationally inefficient and resulting in slow convergence to an optimal path [169]. Addressing this challenge, the research in [176] introduces Informed RRT*, an extension of RRT*. The algorithm focuses the search on the goal area modeled as a prolate hyperspheroid to achieve fast convergence after the RRT* finds an optimal path. RRT, RRT*, and Informed RRT* serve as foundational frameworks for various extensions and adaptations of the traditional RRT algorithm [177]. The figure 12 illustrates how the RRT algorithm identifies a feasible path from a chosen initial position to a designated target location.

Numerous researchers have devised extensions to the traditional RRT algorithm, including RRT-Connect, RRT-D, and the Multipartite RRT (MP-RRT). The RRT-Connect [179] boasts enhanced exploration capabilities by initializing two trees bidirectionally from the outset of the search process: one from the start point and another from the endpoint. It endeavors to link these trees using a straightforward greedy approach, facilitating expedited path planning compared to traditional RRT and mitigating the risk of local minima. An enhanced version of RRT-Connect, proposed in [180], further accelerates the path planning process by introducing a third node in the configuration space, extending the greedy search with a quadruple tree. An Adaptive RRT [181] offers adaptive tree growth in the configuration space, departing from the conventional RRT's full-dimensional planning approach. It selects robot bodies based on path complexity. The Distributed RRT (RRT-D) variant [182] expands RRT by integrating wireless sensor network data, thereby enhancing adaptability in path planning and accommodating dynamic environments. The Multipartite RRT (MP-RRT) [183] addresses planning in dynamic and unmapped environments by reusing branches from prior iterations and biasing the sampling distribution. The simplicity of the RRT makes it a good candidate for planning in dynamic environments [184], [185], as well as when there are kinematic constraints [186].

The RRT* algorithm boasts several variants that offer substantial enhancements. For instance, RRT* -Smart in [172] accelerates convergence toward an optimal solution by reducing execution time. This is accomplished through the integration of path optimization and intelligent sampling techniques. Quick-RRT [187] outperforms traditional RRT* by providing superior initial solutions and convergence rates. It achieves cost-effective paths by increasing the number of parent vertices and selecting vertices within a hypersphere, along with their ancestry, up to a predefined parameter. Additionally, the recently introduced tree-extending algorithm, namely the Fast RRT* (F-RRT*) algorithm, compared with RRT*, RRT*- Smart, and Q-RRT*, demonstrates faster convergence rates and superior initial solutions [173]. RRT* has been implemented in multiple domains, including scenarios where obstacles are dynamic [188], [189] or the target goal is dynamic [190]. The Informed RRT* algorithm offers extensions to address slow convergence. In [191], a wrapping procedure is introduced, enhancing convergence by reducing the size of the hyper-ellipsoid. This is achieved by sampling random path nodes and selecting the path cost of the current best solution. Another extension, Improved Informed-RRT* [192], incorporates an adaptive growth strategy and a variable-weight sampling method within an elliptical region, coupled with trajectory optimization. Compared to Informed RRT*, it achieves a 27.83% reduction in initial path-discovery time and a 16.91% reduction in optimal path-finding time. The RRT algorithm has also been combined with other path-planning algorithms, such as PSO in [193] and PRM in [189], to increase the efficiency and optimality of the algorithm. Among path-planning algorithms, the RRT algorithm is popularly used in dynamic environments, narrow passages, and high-dimensional spaces due to its efficient exploration, quick re-planning, and fast initial solutions [194].

6) OTHER HEURISTIC APPROACHES

This section encompasses various path-planning approaches that have not been previously discussed. Among search-based methods are breadth-first search (BFS), depth-first search (DFS), Uniform Cost Search (UCS), and best-first search (BFS) [195], [196]. Additionally, sample-based methods such as Fast Marching Trees (FMT) exist. Another less common approach is the Vector Field Histogram (VFH) method, utilized for obstacle avoidance and navigation in unknown environments.

Breadth-First Search (BFS) determines the shortest route in a graph or grid by traversing it level by level, exploring all neighboring nodes before moving to the next level until the goal node is reached [195]. The Depth First Search (DFS) traverses a graph or tree structure in a depth-first manner, prioritizing single-branch traversal until reaching a leaf node before backtracking [197]. Uniform Cost Search finds the optimal path between two locations on a grid, considering the associated cost of each edge, and employs other search algorithms to find the optimal path [198].

The Fast Marching Trees algorithm is a real-time planning method that combines local and global path planning, dynamic obstacle avoidance, and efficient tree structure rewiring [199]. It has been utilized for multidimensional optimal motion planning [200], bidirectional optimal motion planning [201], and in conjunction with Support Vector Machines for narrow passage path planning [202].

The Vector Field Histogram (VFH) approach divides the robot's surroundings into sectors, creating a histogram of obstacles in polar coordinates and guiding the robot around obstacles to approach the target point. It finds applications in navigating dense crowds [203] and autonomous navigation [204] in general.

III. ANALYSIS AND DISCUSSION

Based on the literature reviewed in the previous section, the AMR path-planning algorithms can be classified into four types: classical, heuristic, meta-heuristic, and AI-based methods. This section explores recent advancements in classical and heuristic approaches for AMR path planning. Classical techniques, such as cellular decomposition, roadmap methods, and artificial potential fields, excel in structured environments due to their efficiency and simplicity but struggle in complex, dynamic settings (see Table 2 and Fig. 13). To address these limitations, newer sampling- and optimization-based methods have emerged.

Heuristic algorithms, such as Dijkstra and A*, offer faster and more efficient solutions in dynamic environments but may require higher computational power and do not always guarantee optimality. Meta-heuristic approaches, while powerful, demand more significant resources and computational time. Choosing a path-planning algorithm depends on problem complexity, computational capacity, real-time constraints, and the balance between efficiency and accuracy.

In our study of AMR path planning, we reviewed approximately 210 articles from 2017 to 2025, with 80 % sourced from Scopus and 20 % from Web of Science and Google Scholar. Using a structured retrieval strategy, we assessed studies based on abstracts, methodologies, and conclusions, ensuring a thorough analysis of classical and heuristic methods. As shown in Fig. 13, heuristic-based methods are more prevalent in recent works, accounting for approximately 59% of the reviewed studies, while classical approaches represent 41%. This reflects the growing adoption of adaptive and flexible planning techniques, which are better suited to handle complex or uncertain environments.

Figures 13, 14, and 15 collectively highlight key patterns in the validation practices and method preferences observed in recent AMR path planning literature. Figure 13 shows that heuristic-based techniques have become more prominent than classical approaches, reflecting the field's shift toward more flexible, adaptive planning strategies. Figure 14 presents a breakdown of how different algorithms are validated. Based on our data, methods such as Cellular Decomposition (CD)

TABLE 1. An overview of the strengths, advantages, inherent limitations, and challenges associated with classical and heuristic approaches in AMR navigation.

Method	Strengths	Limitations
	Classical Approaches	
CD	Well-suited for a 2D grid-based environment. Guarantees complete coverage.	Difficulty in handling complex and cluttered environments. Limited adaptability to dynamic environments.
RA	Works well for path planning in large-scale environments. Provides global path planning solutions.	Requires substantial pre-processing and memory. Difficulty in handling dynamic environments and obstacles.
APF	Flexible and adaptable to various scenarios and constraints. Can handle dynamic environments with moving obstacles.	Tuning the potential functions can be a challenging task. Local minima and path oscillations can occur.
	Heuristic Approaches	
Dijkstra	Guarantees optimal pathfinding. Suitable for 2D and 3D environments with varying obstacles.	Computationally expensive for large-scale graphs. Not suitable for dynamic environments or real-time applications.
A*	It is an efficient and widely used heuristic algorithm that provides optimal or near-optimal solutions.	The quality of the heuristic function affects performance and optimality. It can be sensitive to the choice of heuristics and graph representation.
D*	Handles dynamic environments with changing costs or graphs; incremental updates enable efficient re-planning.	Complexity can be higher compared to other algorithms. Performance depends on the frequency and magnitude of environmental changes.
Theta*	Enhances A* by eliminating unnecessary nodes and maintains optimality with respect to the original graph.	Requires additional computational overhead for edge collision checking. The density of obstacles can influence path quality.
RRT	Rapidly explores the configuration space, suitable for complex and cluttered environments.	Limited to single-query tasks, may not guarantee optimal solutions, and is susceptible to local minima.

and Theta* are primarily evaluated in simulation environments, with very few implementations tested on physical robotic platforms. This trend is reflected in Figure 15, which shows that 67 % of the surveyed studies rely solely on simulation, while only 3 % perform exclusive real-world testing, and 30 % combine both. These statistics reveal a significant gap in validation between theoretical development and practical deployment, particularly in dynamic environments. Addressing this gap is crucial for enhancing the robustness, reproducibility, and practical applicability of future AMR path planning systems.

Classical and heuristic techniques effectively navigate familiar environments but face challenges in highly dynamic scenarios. The APF and RA techniques are widely used due to their reactive navigation capabilities, ease of implementation, and real-time performance, particularly in structured indoor environments. However, they struggle with scalability in unpredictable environments. Among classical methods, APF remains the most popular for dynamic path organization (Table 2, Fig. 13). Our research review indicates that APF is a powerful, relatively simple, and often straightforward method. However, it does have limitations, including its susceptibility to getting trapped in local minima, which can hinder the robot's progress towards the goal. Moreover, it can become computationally demanding in environments with numerous obstacles, resulting in slowdowns during the path-planning process.

Heuristic approaches, such as D* and Theta*, adapt to environmental changes, making them ideal for dynamic settings. However, their high computational costs limit their efficiency in static environments where re-planning is unnecessary. RRT effectively explores unpredictable spaces but may require numerous iterations for optimal convergence. Notably, dynamic A* (D*), among heuristic methods, is the preferred choice for AMR path planning in continuously changing environments (See Table 2).

Table 1 summarizes the strengths and limitations of classical and heuristic AMR path-planning methods. Table 2 further evaluates their performance in static and dynamic environments, target tracking, empirical validation, and adaptability. While classical and heuristic strategies provide fundamental solutions, they are less flexible in dynamic settings than meta-heuristic and AI-based techniques (see Part 2).

Research on AMR navigation in dynamic environments has increased, with heuristic improvements, such as D* and Theta*, offering promising solutions (Table 2). Classical and heuristic methods remain effective for static or semi-dynamic environments, but researchers continue to enhance them by integrating advanced approaches. Further details on these advancements are discussed in later sections.

Path-planning methods can also be compared using metrics such as computational time, path smoothness, search time, success rate, collision rate, and path length. Studies like [208] found that RRT outperformed RA in search time, while A* and RA yielded shorter paths. Similarly, [209] showed A* outperforming APF in path smoothness and length. In [210], A* and Dijkstra generated the shortest paths, whereas RRT excelled in computation time. These results highlight the trade-offs between efficiency and optimality in different planning techniques.

TABLE 2. A method of comparison between AMR path planning techniques based on some identified parameters.

Method	Article	Year	Kinematic Model	Static obstacle	Dynamic Obstacles	Dynamic targets	Hybrid Approach	Simulation	Real Experiments	Search Space	
										2D	3D
CD	[51]	2018		✓	✓		✓	✓		✓	
	[58]	2017		✓			✓	✓		✓	
	[61]	2017		✓	✓		✓	✓		✓	
	[62]	2019		✓				✓		✓	
	[59]	2021		✓				✓		✓	
	[60]	2021		✓				✓	✓	✓	
	[42]	2022	✓	✓			✓	✓		✓	
	[63]	2021		✓			✓	✓			✓
	[64]	2023	✓	✓				✓		✓	
	[66]	2024		✓			✓	✓		✓	✓
	[67]	2024		✓			✓	✓		✓	
	[65]	2025		✓				✓	✓	✓	
RA	[73]	2017		✓					✓	✓	
	[74]	2019	✓	✓	✓		✓	✓		✓	
	[78]	2021	✓	✓			✓	✓		✓	
	[79]	2021		✓				✓		✓	
	[80]	2022		✓	✓		✓	✓	✓	✓	
	[76]	2022		✓			✓	✓		✓	
	[70]	2025		✓				✓		✓	
APF	[88]	2019	✓	✓	✓			✓	✓	✓	
	[89]	2019	✓	✓	✓			✓	✓	✓	
	[90]	2020	✓	✓				✓	✓	✓	
	[205]	2020		✓	✓			✓		✓	
	[91]	2021		✓	✓			✓		✓	
	[93]	2021		✓	✓		✓	✓	✓	✓	
	[206]	2021		✓			✓	✓	✓	✓	
	[207]	2022		✓			✓	✓		✓	
	[95]	2022		✓	✓		✓	✓		✓	
	[92]	2023	✓	✓	✓			✓	✓	✓	
	[96]	2023		✓			✓	✓		✓	
	[98]	2024	✓	✓				✓	✓	✓	
	[99]	2024	✓	✓	✓	✓	✓	✓	✓	✓	
	[100]	2024		✓			✓	✓	✓	✓	
	[101]	2024		✓	✓		✓	✓		✓	
	[94]	2025	✓	✓	✓		✓	✓		✓	
	[97]	2025		✓			✓	✓		✓	
Dijkstra	[7]	2021	✓	✓	✓			✓	✓	✓	
	[107]	2020		✓				✓		✓	
	[108]	2018		✓				✓		✓	
	[109]	2021		✓				✓		✓	
	[111]	2021		✓	✓		✓	✓		✓	
	[110]	2022	✓	✓	✓			✓	✓	✓	
	[112]	2023	✓	✓			✓	✓	✓	✓	
	[113]	2023	✓	✓			✓	✓	✓	✓	
	[114]	2023		✓			✓	✓		✓	
	[116]	2024		✓			✓	✓		✓	
	[117]	2024		✓				✓		✓	
	[118]	2024		✓					✓	✓	
	[115]	2025	✓	✓	✓		✓	✓		✓	
A*	[126]	2022		✓				✓		✓	
	[127]	2014		✓	✓			✓	✓	✓	
	[124]	2019		✓			✓	✓		✓	
	[128]	2016		✓				✓		✓	
	[129]	2022		✓	✓	✓	✓	✓		✓	
	[133]	2022		✓			✓	✓		✓	
	[125]	2023		✓			✓	✓			✓
	[135]	2024		✓				✓		✓	
	[130]	2022		✓	✓		✓	✓		✓	
	[136]	2022	✓	✓			✓	✓	✓	✓	
	[137]	2022		✓			✓	✓		✓	
	[131]	2024	✓	✓			✓	✓	✓	✓	
	[132]	2025		✓	✓		✓	✓		✓	
	[138]	2024		✓				✓	✓	✓	
D*	[141]	2023		✓	✓		✓	✓		✓	
	[142]	2021		✓	✓			✓	✓		✓
	[146]	2018		✓	✓		✓	✓		✓	
	[147]	2019		✓	✓		✓	✓		✓	
	[148]	2022			✓	✓	✓	✓			✓
	[150]	2022	✓	✓	✓	✓		✓	✓	✓	
	[151]	2024		✓				✓		✓	

TABLE 3. A method of comparison between AMR path planning techniques based on some identified parameters.

Method	Article	Year	Kinematic Model	Static obstacle	Dynamic Obstacles	Dynamic targets	Hybrid Approach	Simulation	Real Experiments	Search Space	
										2D	3D
Theta*	[160]	2022		✓		✓		✓			✓
	[161]	2021	✓		✓		✓		✓	✓	
	[162]	2022		✓				✓		✓	
	[163]	2023		✓			✓	✓			✓
	[164]	2023	✓	✓			✓	✓		✓	
	[165]	2024		✓				✓		✓	
	[166]	2024		✓			✓	✓			✓
RRT	[170]	2020		✓			✓	✓		✓	
	[180]	2021		✓				✓	✓	✓	
	[174]	2021		✓				✓		✓	
	[185]	2021			✓			✓		✓	
	[186]	2019	✓	✓				✓		✓	
	[184]	2023		✓	✓		✓	✓		✓	
	[190]	2019		✓	✓	✓		✓			✓
	[192]	2023		✓				✓		✓	
	[189]	2025		✓	✓		✓	✓	✓	✓	
	[194]	2025		✓				✓	✓	✓	
	[193]	2025		✓				✓	✓	✓	

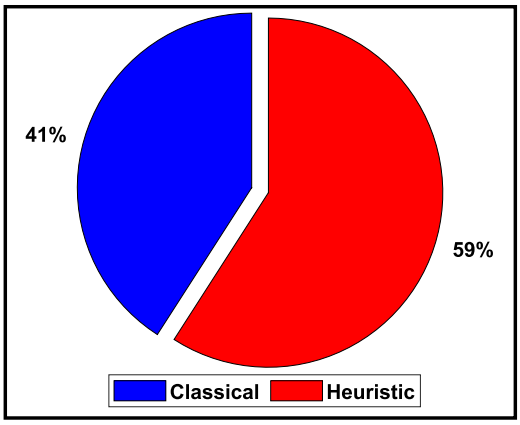


FIGURE 13. Statistical analysis of various path planning algorithms for dynamic environments, based on a review of approximately 210 papers published between 2017 and 2025.

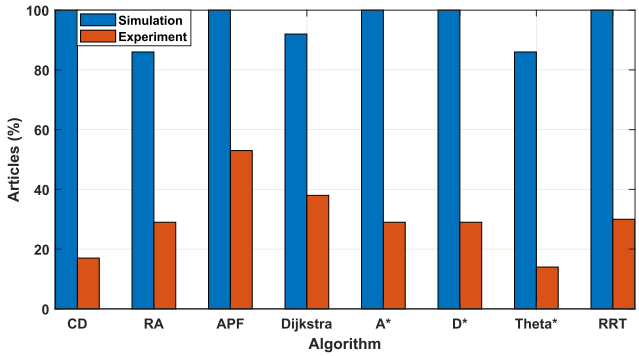


FIGURE 14. Validation methodology distribution for AMR path planning algorithms in this literature review. The figure presents the proportion of studies utilizing simulation-based validation versus real-world experimentation for each algorithm.

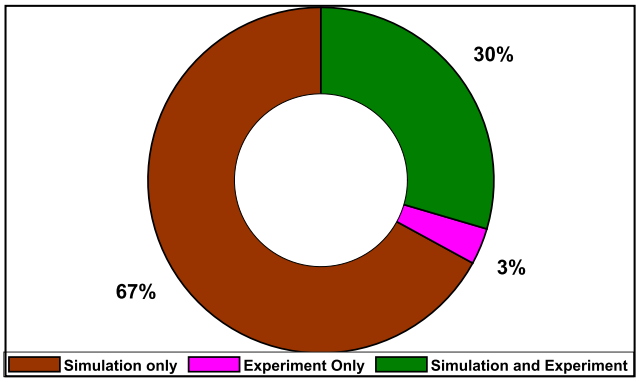
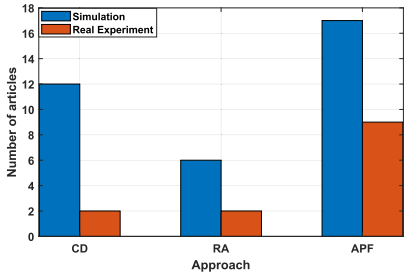


FIGURE 15. Approach for validating the planning algorithms discussed in this literature review.

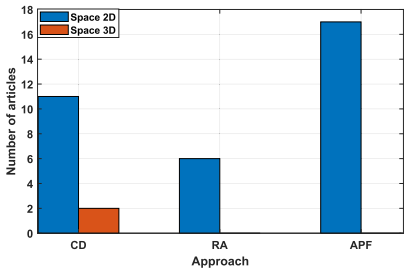
In particular, we analyze the distribution and trends in publications on navigation approaches, categorized by dimensional space (2D vs. 3D), testing environment (simulation vs. real-world), and obstacle dynamics (static vs. dynamic). Both traditional and heuristic methods are examined.

Figures 16a, 16b, and 16c present traditional approaches, while Figs. 17a, 17b, and 17c illustrate heuristic approaches. These figures encompass variations in simulation versus real-world testing, 2D versus 3D environments, and static versus dynamic obstacles.

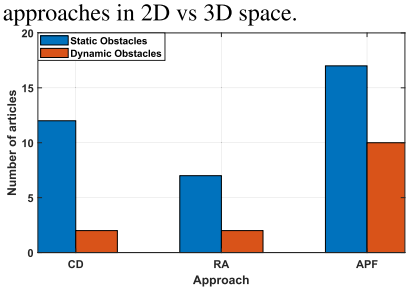
A notable trend in the literature is the increasing focus on early and ongoing research in 2D environments. This focus is mainly due to the reduced complexity of 2D settings, which makes them more suitable for initial experimentation and exploration. Heuristic algorithms, in particular, demonstrate strong performance in 2D scenarios, excelling in cluttered environments and tasks involving obstacle avoidance. Simulation-based testing dominates the field in both traditional and heuristic methods. Simulations offer a safe, reproducible, and flexible platform for evaluating navigation algorithms, enabling rapid development and testing. As a result, the vast majority of studies involving heuristic approaches are conducted in simulated environments.



(a) Distribution of traditional navigation approaches in Simulation vs Real experiment.

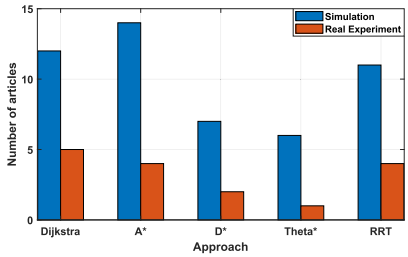


(b) Distribution of traditional navigation approaches in 2D vs 3D space.

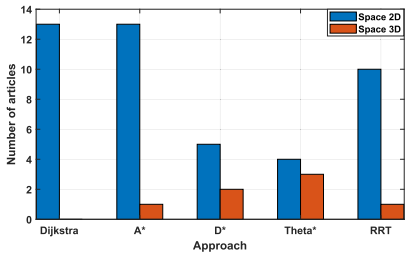


(c) Distribution of traditional navigation approaches in static and dynamic obstacles.

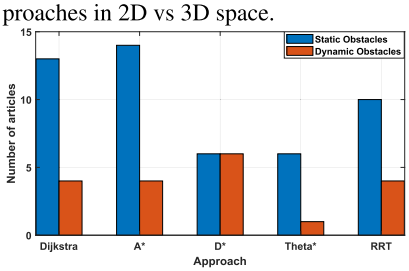
FIGURE 16. Overview of traditional navigation approaches.



(a) Distribution of heuristic navigation approaches in simulation vs. real experiments.



(b) Distribution of heuristic navigation approaches in 2D vs 3D space.



(c) Distribution of heuristic navigation approaches in static and dynamic obstacles.

FIGURE 17. Overview of heuristic navigation approaches.

While recent research trends indicate an increasing interest in dynamic obstacle handling, particularly with heuristic methods such as D*, the majority of studies still focus on static obstacle scenarios. This preference is primarily due to the reduced complexity and ease of reproducibility in such environments, particularly during early-stage algorithm testing.

These distributional insights further support the need for future research to focus more on real-world experimentation, three-dimensional navigation, and dynamic obstacle scenarios to enhance the robustness and generalizability of AMR path planning solutions.

IV. CHALLENGES AND OPPORTUNITIES IN AMR PATH PLANNING

A. CHALLENGES

Effective path planning is crucial for AMRs to navigate dynamic and diverse environments. However, developing robust path-planning algorithms poses several challenges that require innovative solutions while creating new opportunities for advancement.

1) ENVIRONMENTAL CHALLENGES

- Dynamic and Uncertain Environments: AMR operates in environments with unpredictable obstacles, dynamic terrains, and moving objects. AMRs must adapt to dynamically changing environments with moving obstacles and unexpected terrain variations.
- Limited Sensor Range and Accuracy: Sensors such as LiDAR, RGB-D cameras, and GPS provide real-time data; however, sensor noise and inaccuracies can lead to localization errors, which in turn affect the reliability of path planning.

2) COMPUTATIONAL CHALLENGES

- High computational complexity: Real-time path planning involves determining the optimal route while minimizing costs, including distance, time, and energy. Large-scale environments increase computation time and require substantial computational power, especially in multi-robot systems.

- Local Minima in heuristic algorithms: Traditional algorithms fail in environments with narrow passages or obstacles forming concave regions. They suffer from local minima, leading to suboptimal paths or getting stuck, leading to inefficient or infeasible paths.
- Energy efficiency and resource constraints: Optimizing path planning for energy consumption is crucial, especially for battery-powered robots. Therefore, path planning must consider battery limitations for extended autonomous mobile robot (AMR) operation.

B. OPPORTUNITIES

Despite the inherent limitations of existing algorithms, their fundamental mathematical principles remain accepted. Therefore, optimizing path-planning algorithms while considering these constraints is essential.

1) OPTIMIZING THE PERFORMANCE OF EXISTING ALGORITHMS

Future research should explore and integrate advanced mathematical theories to enhance these algorithms. For instance, improving the evaluation function of the DWA algorithm and refining the cost functions of the A* and D* algorithms are promising directions for further development.

2) HYBRID APPROACHES

Future research should explore hybrid strategies that leverage the strengths of different path-planning algorithms to overcome their limitations. For instance,

- Integrating the DWA with global planners such as Dijkstra, D*, A*, Theta*, or RRT could enhance real-time adaptability while maintaining efficient global path planning. Hybrid models could utilize Dijkstra, A*, or Theta* for initial long-range path generation and rely on DWA or RRT for dynamic local obstacle avoidance, ensuring seamless and efficient navigation.
- Hybrid models could utilize cell decomposition and RRT for dynamic local obstacle avoidance, ensuring seamless and efficient navigation while reducing computational complexity.
- Incorporating machine learning techniques into these hybrid models could enhance decision-making by dynamically adjusting evaluation and cost functions based on environmental conditions. For example, reinforcement learning could optimize the transition between global and local planning strategies, while deep learning could refine the heuristic functions in A* or Theta* to improve computational efficiency.

By combining heuristic search methods with real-time reactive algorithms, hybrid approaches can create more robust, efficient, and adaptive path-planning solutions, ultimately leading to safer and more reliable autonomous navigation.

TABLE 4. List of abbreviations.

Acronym	Meaning
AMR	Autonomous Mobile Robots
WMR	Wheeled Mobile Robot
CD	Cell Decomposition
DA	Dijkstra Algorithm
A*	A Star
RA	Road-map Approach
VD	Voronoi Diagram
VG	Visibility Graph
APF	Artificial Potential Field
PRM	Probability Road-map
RRT	Rapid-exploring Random Trees
D*	Dynamic A*
PSO	Particle Swarm Optimization
BSA	Beetle Search Algorithm
GWO	Grey Wolf Optimizer
LS	Local Search
BA	Bat Algorithm
ACO	Ant Colony Optimization
ODA	Obstacle Detection and Avoidance
SA	Simulated Annealing
DWA	Dynamic Window Approach
RL	Reinforcement Learning
ROS	Robot Operating System

3) ENERGY-AWARE HEURISTICS FOR PATH PLANNING APPROACH

Path planning for AMRs traditionally focuses on finding the shortest or fastest route. However, in energy-constrained systems, energy efficiency is a crucial consideration. Energy-aware path planning modifies classical algorithms, such as A* and D*, to incorporate energy consumption as a key optimization metric. These approaches focus on minimizing distance changes but do not explicitly account for energy costs. To account for energy consumption, the cost functions can be modified to factor in energy-related metrics.

V. CONCLUSION AND FUTURE PROSPECTS

Autonomous mobile robot (AMR) path planning is a critical research area that has gained significant attention in recent years due to the growing demand for efficient and safe navigation of robots in complex environments. This review article provides a comprehensive overview of the current state of AMR path planning, encompassing various techniques and algorithms that have been developed. We have classified these methods into different categories, discussed their strengths and limitations, and provided examples of their applications in multiple domains. We also highlighted some of the recent trends and challenges within the field and discussed the potential impact of path planning on the overall performance and safety of autonomous mobile robots (AMRs).

Our research survey has revealed various approaches to AMR path planning, each with advantages and limitations. Classical, heuristic, meta-heuristic, and learning/reasoning-based methods are the most commonly employed techniques, and their performance has been evaluated in various domains. However, several research gaps and challenges still need to be addressed, including the development of more robust and adaptive algorithms, the integration of sensor fusion and

perception techniques, and the incorporation of human factors into the planning process. This review is intended to serve as a resource for researchers and practitioners, supporting further innovation in AMR path planning and encouraging the development of practical, real-world solutions that enhance the autonomy and safety of mobile robotic systems.

DATA AVAILABILITY STATEMENT

All data supporting the findings of this study are included within the article or cited in the referenced materials.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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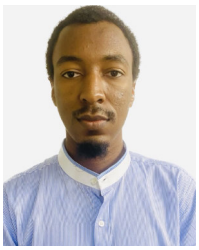

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